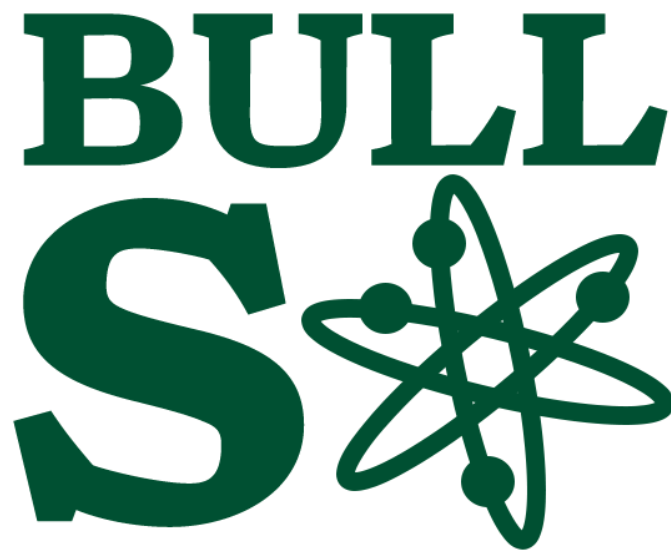


BullSO 2026 Invitational
Meteorology B



Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name: _____

Participant Names: _____

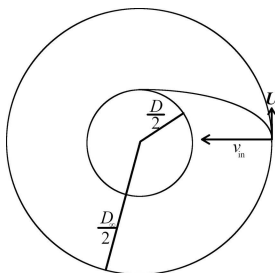
Total Score: _____/60

Rank: _____

Part 1: MCQs

1. During a strong El Niño event, sea surface temperatures in the eastern equatorial Pacific become significantly warmer than average. Which combination of changes best explains the resulting global climate and ecological impacts?
 - a. Weakened trade winds reduce coastal upwelling in the eastern Pacific, causing decreased nutrient availability for phytoplankton and increased rainfall along the western coast of South America.
 - b. Strengthened trade winds increase upwelling off the coast of South America, leading to higher marine productivity and increased rainfall in Australia.
 - c. Enhanced Walker circulation shifts warm surface waters westward, intensifying monsoon systems in Southeast Asia and increasing fish populations near Peru.
 - d. Increased atmospheric pressure over the eastern Pacific strengthens subtropical jet streams, resulting in colder ocean temperatures and reduced precipitation in South America.

2. A weather station located in a narrow valley reports clear skies and calm winds overnight, followed by rapid cooling of the ground after sunset. By early morning, dense fog forms near the valley floor, while higher elevations remain clear. Which type of fog is most likely responsible, and what is the primary mechanism of its formation?
 - a. Advection fog formed when warm, moist air moves horizontally over a colder surface
 - b. Valley fog formed by cold, dense air draining downslope and becoming trapped, allowing saturation to occur
 - c. Steam fog formed when cold air moves over warmer water, causing rapid evaporation and condensation
 - d. Radiation fog formed by nocturnal radiative cooling of the ground, chilling the air near the surface to its dew point



3. The above image is of a dust devil which is expressed in a math context. The innermost circle, with radius $D/2$, represents the convective core of the dust devil, while the outermost circle, with radius $D_{\infty}/2$, represents the diameter of influence. The wind component perpendicular to the dust devil is approximated by

$U = \alpha(D\infty/2)$ where α is lateral wind shear and radial inflow will be given by V_{in} ,

Which of the following changes would most likely result in a **stronger and more organized dust devil**, assuming surface heating remains constant?

- a. Decreasing α , which reduces perpendicular wind speed and limits rotational inflow
- b. Increasing $D\infty$, which increases perpendicular wind speed and enhances inward spiraling motion
- c. Decreasing V_{in} , which allows angular momentum to concentrate near the core
- d. Reducing the convective cell radius $D/2$, which decreases vertical motion and weakens rotation

4. The Coriolis effect affects the motion of air and water across the surface of the Earth via Earth's rotation, which of the following would explain why the Coriolis is weak near the equator yet strongest near the poles?

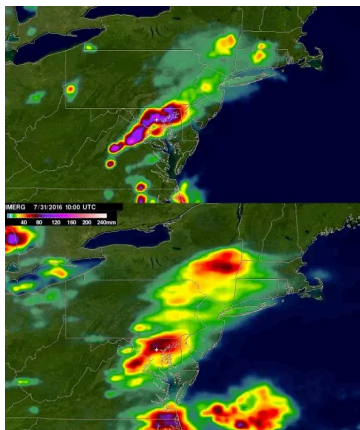
a. Earth's rotational speed increases toward the poles, causing moving air and water to experience greater apparent deflection at higher latitudes.

b. The angular velocity of Earth's rotation changes with latitude, producing stronger deflection of moving objects near the poles.

c. The linear (tangential) velocity of Earth's surface decreases from the equator toward the poles, increasing the relative deflection of moving air and water.

d. Differences in Earth's rotation cause air and water to move faster at higher latitudes, increasing curvature in their paths.

5- 8 questions are based on the image below:



(Image of a storm forecast of Maryland)

5. The images shows two precipitation maps of the northeastern United States at 00:00 UTC (top) and 10:00 UTC (bottom). Colors range from green to red/purple. Which conclusion is best supported by the change between the two images?
- a. Precipitation intensity decreases over time as the storm system weakens.
 - b. The storm system expands and intensifies, with heavier precipitation covering a larger area by 10:00 UTC.
 - c. Precipitation remains stationary, indicating a stalled frontal boundary.
 - d. The storm dissipates over land and reforms over the Atlantic Ocean.
6. Based on the spatial distribution of precipitation between 00:00 UTC and 10:00 UTC, the storm system is most likely moving:
- a. Westward due to prevailing trade winds
 - b. Eastward into the Atlantic Ocean
 - c. Northeastward along the U.S. East Coast
 - d. Southward toward the Gulf of Mexico
7. Given that the second image shows increased precipitation coverage after 10 hours, which inference is most reasonable?
- a. The system is losing moisture as it moves inland
 - b. Atmospheric instability is decreasing
 - c. Continued lifting and moisture supply are sustaining the storm
 - d. The Coriolis effect is weakening the system near the equator
8. Which of the following technology could they have used to construct the imaging given in this image?
- a. Ground-based weather stations only
 - b. Doppler radar measuring wind speed
 - c. Multiple satellites measuring microwave and infrared signals
 - d. Weather balloons measuring air pressure
9. A cold front moves through a region, causing rising air to cool rapidly. Which type of cloud is most likely to form and potentially produce precipitation?
- a. Cirrus
 - b. Cumulonimbus
 - c. Stratus
 - d. Altocumulus



10. Identify the cloud formation based on the image above?

- a. Cirrostratus
- b. Altocumulus
- c. Altostratus
- d. Altocumulous

11. A high, wispy clouds composed mostly of ice crystals are typically found at altitudes above 6 km. Which cloud type does this describe?

- a. Cirrus
- b. Nimbostratus
- c. Stratocumulus
- d. Altostratus

12. A coastal city experiences fog in the early morning during late spring. Warm, moist air from the ocean moves inland and passes over the cooler land surface, causing condensation near the ground. Which type of fog is most likely forming, and why?

- a. Radiation fog — caused by nocturnal cooling of the ground, chilling the air near the surface
- b. Advection fog — formed when warm, moist air moves horizontally over a colder surface
- c. Steam fog — formed when cold air moves over warmer water, causing rapid evaporation
- d. Valley fog — caused by cold, dense air draining into low-lying areas and pooling

13. Scientists discovered a significant depletion of ozone over Antarctica, commonly called the “ozone hole.” Which of the following best explains the cause of this phenomenon?

- a. Increased solar activity producing more ultraviolet radiation, which directly destroys ozone molecules
- b. Human-made chemicals such as CFCs releasing chlorine in the stratosphere, which catalytically destroys ozone, especially in polar regions
- c. Volcanic eruptions injecting large amounts of carbon dioxide, which reacts with ozone in the stratosphere
- d. Excessive greenhouse gases warming the lower atmosphere, causing ozone near the surface to break down

14. Most weather events, including clouds, rain, and storms, occur in which layer of the atmosphere, and why?

- a. Troposphere — it contains most of the atmosphere's water vapor and is where temperature decreases with altitude, allowing convection and weather
- b. Stratosphere — it contains the ozone layer and temperature increases with altitude
- c. Mesosphere — it is very cold and burns up meteors entering the atmosphere
- d. Thermosphere — it contains the ionosphere, which reflects radio waves

15. The specific heat of water is $c=4.18 \text{ J/g}^\circ\text{C}$. If 500 g of water is heated from 20°C to 50°C , how much heat energy Q is absorbed by the water?

- a. 5,400 J
- b. 125,100 J
- c. 42,900 J
- d. 62,700 J

16. A warm, moist air mass rises over a mountain range, cools adiabatically, and begins forming clouds. Which combination of processes best explains how precipitation develops in this scenario?

- a. Air cools to its dew point, condensation forms droplets on cloud condensation nuclei; droplets collide and coalesce until heavy enough to fall as rain; some water may freeze and melt on the way down.
- b. Air warms as it rises, causing ice crystals to sublime; sublimation increases droplet size, producing rain at the surface.
- c. Water vapor condenses only after droplets reach the ground, producing precipitation directly from the surface.
- d. Air becomes saturated, and dry air currents absorb excess water vapor, preventing cloud and rain formation.

17. Which property of water allows oceans to moderate coastal temperatures?

- a. High specific heat
- b. Low density
- c. Low latent heat
- d. Rapid freezing

18. Which of the following is an accurate description of Alberta's Clipper?

- a. Slow-moving, moisture-rich system from the Gulf of Mexico producing heavy snow and ice
- b. Low-pressure system forming over the Pacific Ocean that intensifies into a severe winter storm over the northern U.S.
- c. High-pressure system developing in the central Plains, bringing calm weather and clear skies
- d. Fast-moving, low-pressure system from western Canada, limited in moisture, bringing light snow, gusty winds, and a sharp drop in temperature

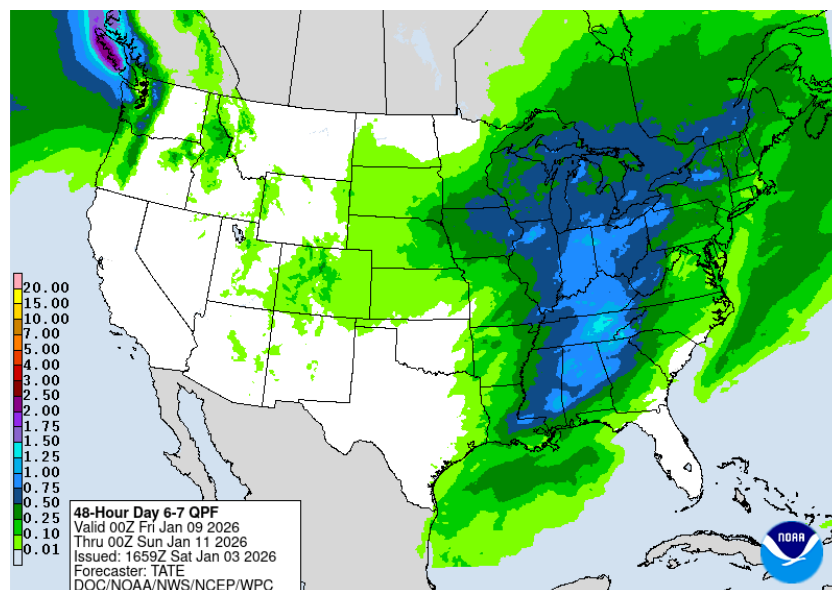
19. On a hot summer day, a driver sees what appears to be water on the road ahead, but when they reach it, there is nothing there. Which explanation best matches this phenomenon?

- a. Light is reflected off distant clouds, creating the illusion of water.
- b. Light bends as it passes through layers of air at different temperatures, creating an image of the sky on the ground.
- c. The appearance of water is an optical illusion caused by the refraction of light through hot air near the surface
- d. Dust particles scatter sunlight, producing a shimmering effect that looks like water.

20. Which instrument measures humidity in the atmosphere?

- a Barometer
- b. Anemometer
- c. Hygrometer
- d. Thermometer

Part 2: Short answer responses



(Questions 21- 26 will be based on the image shown above)

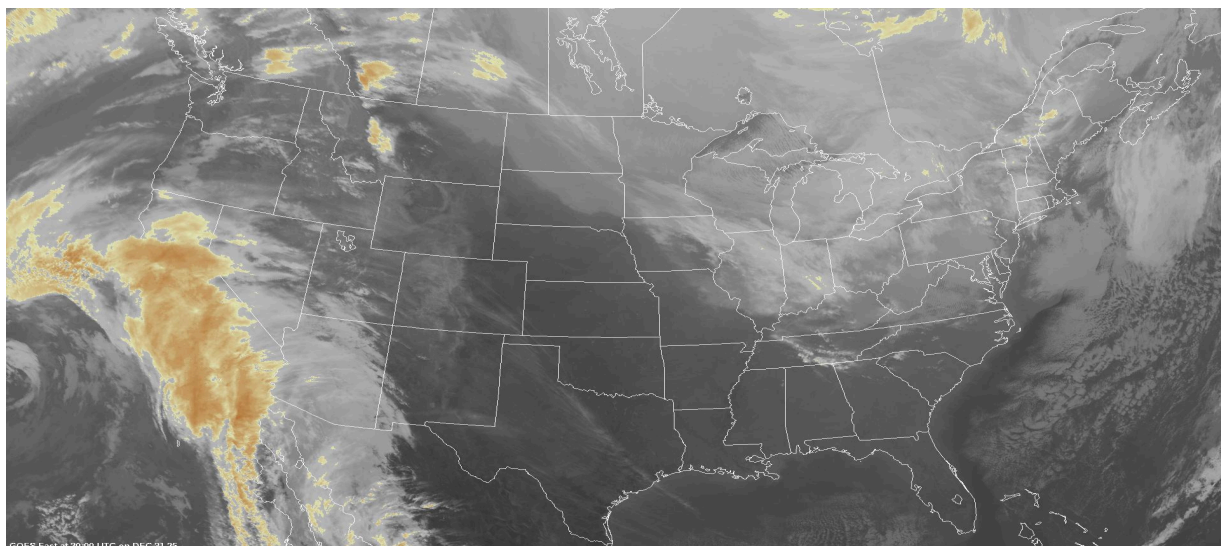
21. Examine the forecasted precipitation bands. How might the orientation and shape of these bands relate to the position of the upper-level jet stream? Explain your reasoning

22. Explain why a forecast map like this might show precipitation even in areas without obvious cloud cover on satellite images.

23. With the following information of the green shading represents 0.5–1.0 inch of rain, the yellow shading represents 1.0–1.5 inches, and the red shading represents 2.0–3.0 inches of precipitation. Assume the total area covered by green is 20,000 km², yellow is 10,000 km², and red is 5,000 km². Estimate the total volume of precipitation (in cubic meters) falling over the entire forecast area, assuming 1 inch = 0.0254 meters. Show your work.

24. Based on the forecasted precipitation pattern, which areas might experience localized flooding or saturated soils, even though this is a non-severe storm? Justify your answer using map colors and atmospheric processes.

25. Critically evaluate one limitation of using precipitation forecast maps like this for predicting real-time weather impacts. How could forecasters compensate for this limitation?



(Questions 26-30 will be based on the image above of the map)

26. Identify a region on the image where cloud tops are likely highest and coldest, and explain how infrared satellite data indicates this.

27. Explain how this IR satellite image could be used, in combination with surface weather maps, to identify the location of a frontal boundary?

28. Using your understanding of how IR imagery depicts temperature, infer whether this snapshot was taken during the day or night and justify your reasoning.

29. Infrared imagery can detect clouds both day and night. Describe how the IR color data could be combined with surface temperature measurements to predict fog formation overnight.

30. Some cloud patterns on the image appear as elongated bands. Describe what atmospheric phenomenon could be causing these bands and how this might relate to the jet stream or surface fronts.